Arkaraj Mukherjee
Worksheet 12

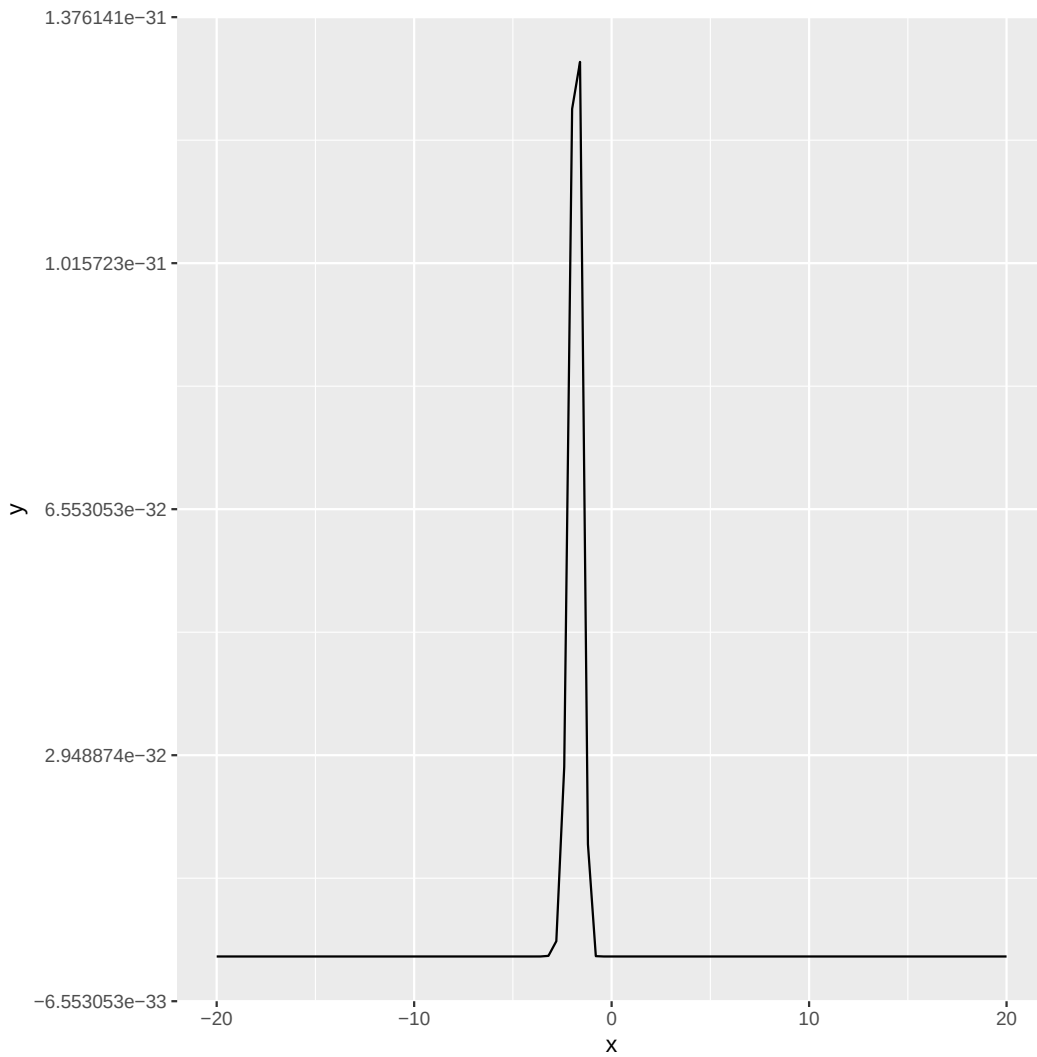
```
library("tidyverse");


## - Attaching core tidyverse packages ------------------- tidyverse 2.0.0
-
## v dplyr     1.2.0     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## - Conflicts ------------------------------ tidyverse_conflicts() -
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force
all conflicts to become errors
```

1. (a) We have that

$$L(\beta_1) = \prod_{i=1}^{15} \prod_{j=1}^{15} \frac{e^{\beta_1 (x_i - x_j)^2}}{1 + e^{\beta_1 (x_i - x_j)^2}}$$

(b)
```
x = c(11,9,12,9,10,11,7,9,10,9,10,10,9,11,8)
# we make the adjacency matrix first
G = matrix(data=0,nrow=15,ncol=15)
G[1,c(6,9)]=1
G[2,c(8,12)]=1
G[4,10]=1
G[5,9]=1
G[6,14]=1
G[7,15]=1
G[8,c(10,11,15)]=1
G[9,11]=1
G[13,15]=1
G = G + t(G)
Prob = function(x,i,j,beta) {
exp(beta*(x[i]-x[j])^2)/(1+exp(beta*(x[i]-x[j])^2))
}
init_L = function(x,G,beta) {
        P = 1
        for(i in 1:nrow(G)) {
                for(j in 1:ncol(G)) {
```

```
                        if( G[i,j] == 1 ) {P = P * Prob(x,i,j,beta)}
                        if( G[i,j] == 0 ) {P = P * (1 - Prob(x,i,j,beta))}
                }
        }
        P
}
L_1 = function(beta) { init_L(x,G,beta) }
ggplot() +
geom_function(fun = L_1, xlim = c(-20,20))
```

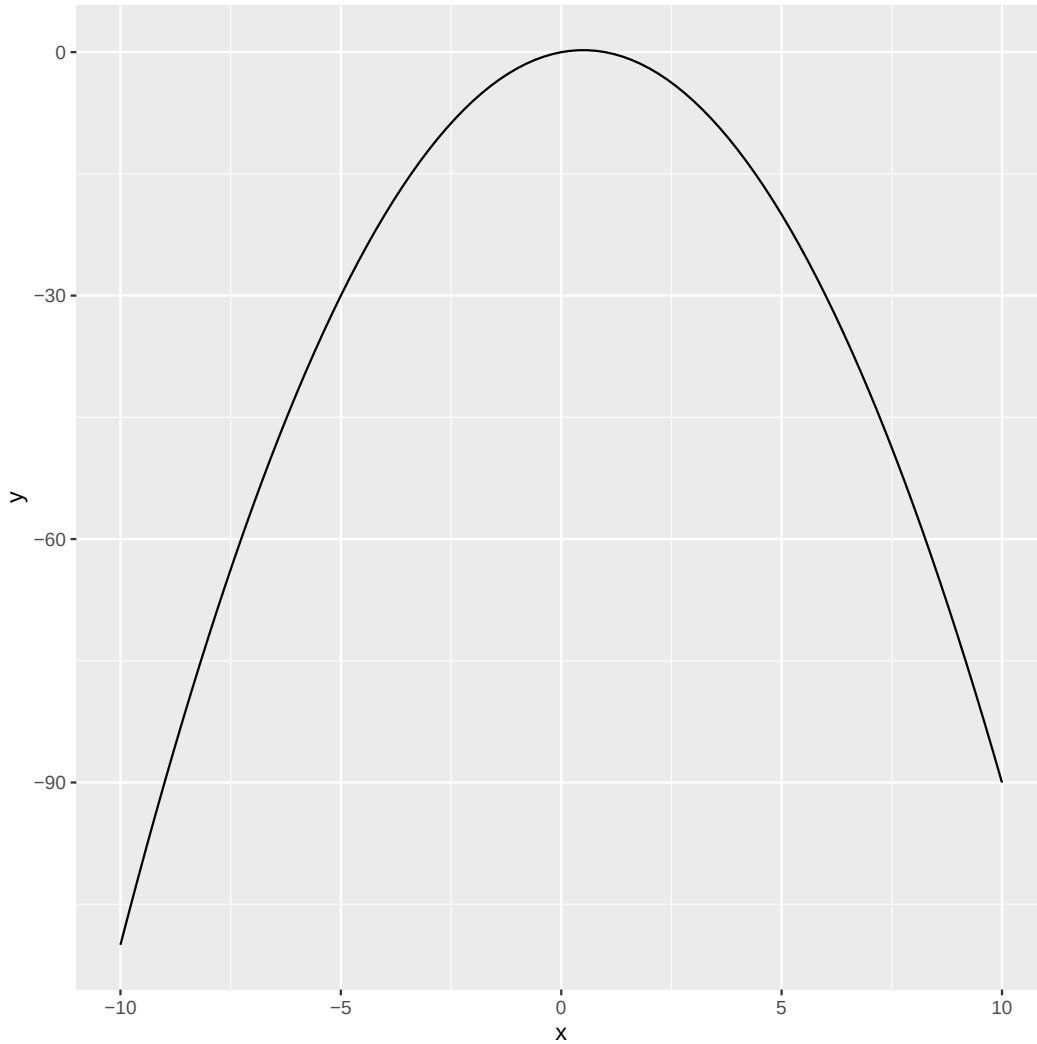



```
#clearly L is C_c(R) and we see that it spikes near 0, in (-10,10)
beta_1 = optimise(L_1, c(-10,10),maximum=TRUE)$maximum
beta_1

## [1] -1.781288
```

This is a good guess by MLE theory

2. (a)

```r
#clearly the function to plot is p(1-p)
f_1 = function(x) x*(1-x)
ggplot()+
geom_function(fun = f_1, xlim = c(-10,10))
```

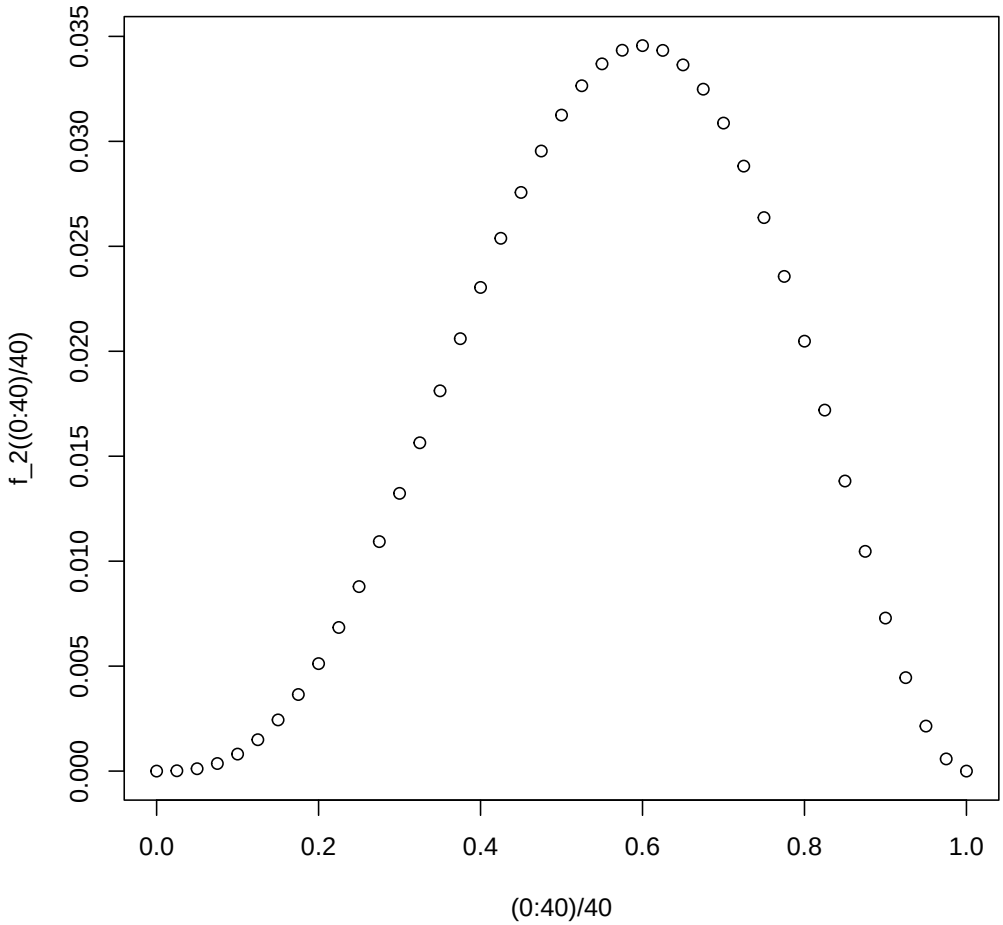



```r
#the maximum is
optimise(f_1, c(-10,10),maximum=TRUE)$maximum

## [1] 0.5
```

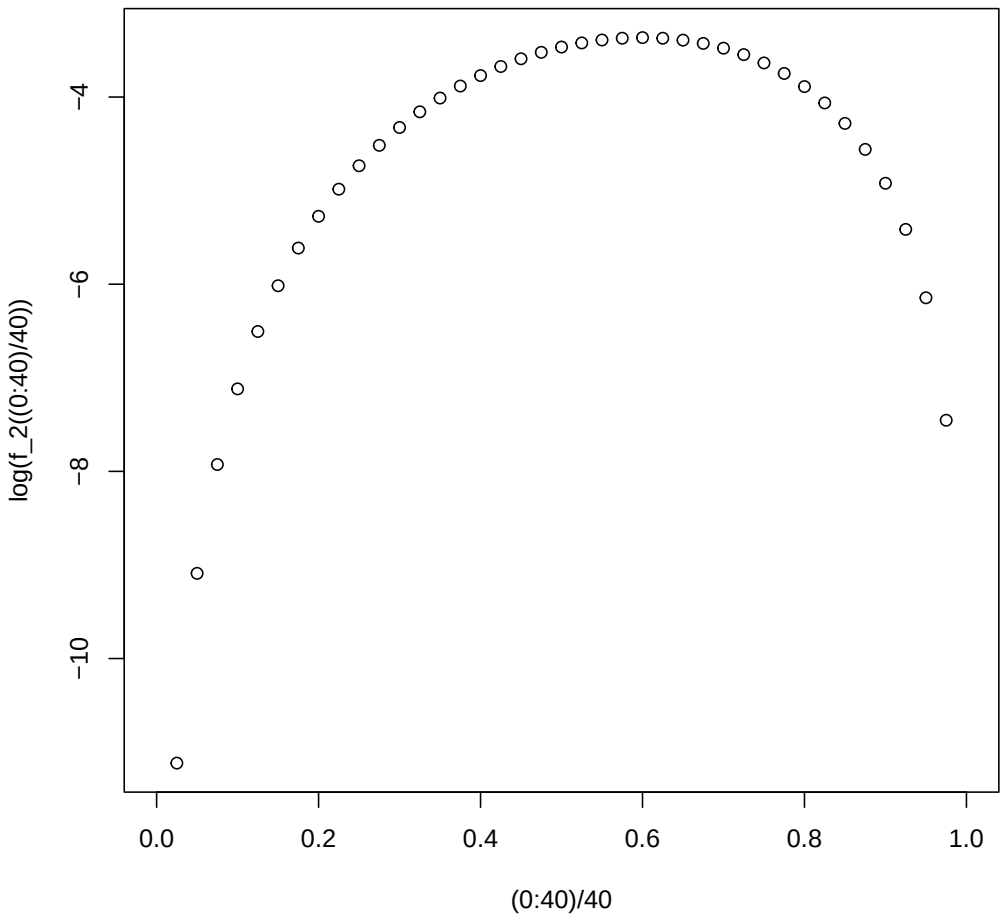
(b) This probability is $p^{\sum_{i=1}^n y_i}(1-p)^{n-\sum_{i=1}^n y_i}$ as $Y_1, \ldots, Y_n \sim$ i.i.d. $\text{Bin}(1, p)$

(c) Here we clearly see that $L(p) = p^3(1-p)^2$

```r
f_2 = function(x) x^3 * (1-x)^2
plot(x=(0:40)/40, y = f_2((0:40)/40))
```




```
plot(x=(0:40)/40, y = log(f_2((0:40)/40)))
```



(d)
```
optimise(f_2, c(0,1), maximum=TRUE)$maximum

## [1] 0.6000006
```

(e) $\hat{p}$ is a good guess for the actual value of p that might result in such a result by MLE theory.

(f)

$$\log \tilde{L}(\beta_0) = \log L(p) = \log\left(p^{\sum y_i}(1-p)^{n-\sum y_i}\right) = \log\left(\left(\frac{e^{\beta_0}}{1+e^{\beta_0}}\right)^{\sum y_i}\left(\frac{1}{1+e^{\beta_0}}\right)^{n-\sum y_i}\right)$$

$$= \log\left(e^{\beta_0 \sum y_i}(1+e^{\beta_0})^{-n}\right) = n\log\left(\frac{1}{1+e^{\beta_0}}\right) + \sum_{i=1}^{n}\beta_0 y_i = \sum_{i=1}^{n}\left(y_i\beta_0 + \log\left(\frac{1}{1+e^{\beta_0}}\right)\right)$$

We will proceed finding the `argmax` by maximising the log variant instead as log is a strictly increasing function on its domain. We will use the double

derivative test

$$\frac{d}{d\beta_0}\log\tilde{L}(\beta_0) = \sum_{i=1}^{n} y_i + n\frac{e^{\beta_0}}{(1+e^{\beta_0})} = 0 \implies \frac{e^{\beta_0}}{(1+e^{\beta_0})} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

$$\frac{d^2}{d\beta_0^2}\log\tilde{L}(\beta_0) = -n\left(\frac{e^{\beta_0}}{(1+e^{\beta_0})^2}\right) < 0$$

so by the double derivative rule there is unique maxima, achieved when the first derivative is 0, solving that linear equation in e^{β_0} and taking log gives,

$$\hat{\beta}_0 = \log\left(\frac{\bar{y}}{1-\bar{y}}\right)$$

after some manipulation, where $\bar{y} = n^{-1}\sum_{i=1}^{n} y_i$

(g) It is $\log((3/5)/(1-(3/5))) = \log((3/5)/(2/5)) = \log(3) - \log(2)$

```
log(1.5)

## [1] 0.4054651
```

(h) From the previous part,a good guess for p here would be $\bar{y} = 3/5$ so the probability for them to to win their next match would be (as the outcomes are independent), $p = 3/5$

3. (a) by the previous problem we see that,

```
beta_0_hat = log((724/2224)/(1-(724/2224)))
beta_0_hat

## [1] -0.728429
```

(b) The probability of survival for any passenger is 724/2224 which is

```
724/2224

## [1] 0.3255396
```

(c) The probability of death approximately $1 - 0.674 = 0.326 < 0.5$ so we can predict that this person likely died.

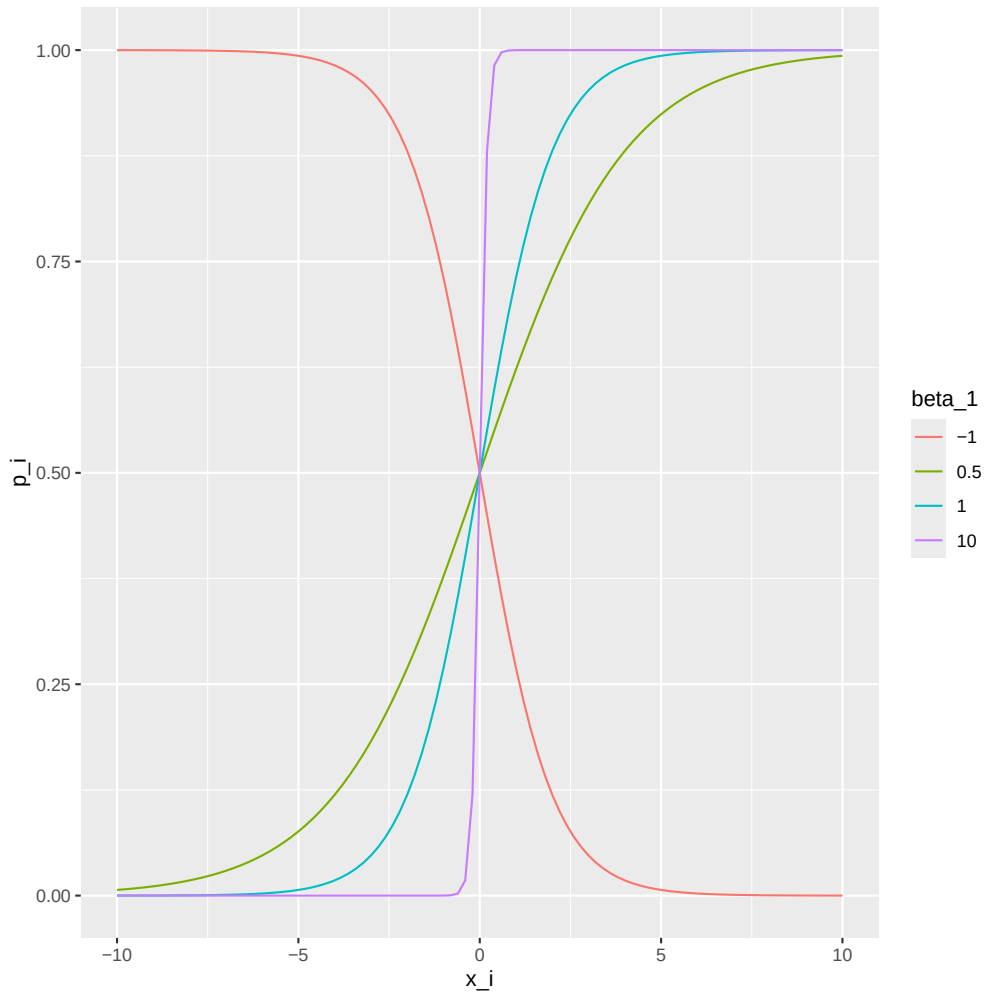
4. (a) The lower levels (more depth) should correspond to lower fairs so we consider functions that map fare to depth and these have to be decreasing. Five examples of functions $f : \mathbb{R} \to [0, 1]$ are:

- $f_1(x) = \frac{e^x}{1+e^x}$
- $f_2(x) = \Phi(x)$standard normal cdf
- $f_3(x) = \frac{1}{\pi}\arctan(x) + \frac{1}{2}$
- $f_4(x) = \frac{1}{2}(1 + \text{erf}(x))$

- $f_5(x) = e^{-x^2}$

(b) i.

```
p_i = function(x, beta_1) { exp(beta_1*x)/(1+exp(beta_1*x)) }
ggplot() +
geom_function(fun = function(x) p_i(x, -1), aes(color = "-1"),
xlim=c(-10,10)) +
geom_function(fun = function(x) p_i(x, 0.5), aes(color = "0.5"),
xlim=c(-10,10)) +
geom_function(fun = function(x) p_i(x, 1), aes(color = "1"),
xlim=c(-10,10)) +
geom_function(fun = function(x) p_i(x, 10), aes(color = "10"),
xlim=c(-10,10)) +
labs(color = "beta_1", x = "x_i", y = "p_i")
```



ii. The function $p_i(x_i)$ is an increasing function of x_i when $\beta_1 > 0$. An increasing function implies that as the fare (x_i) increases, the probability of survival (p_i) also increases. Conversely, a decreasing function (when $\beta_1 < 0$) implies that paying a higher fare corresponds to a lower probability

of survival.

iii. When $x_i = 0$, $p_i = \frac{e^0}{1+e^0} = \frac{1}{2} = 0.5$. This seems too high for the Titanic data. Since the overall baseline survival rate was much lower (approximately 32.55%), a fare of 0 (representing the poorest passengers or crew members) should correspond to a survival probability significantly lower than the average.

iv. We can modify the function by adding an intercept term β_0 to the exponent:

$$p_i(x_i) = \frac{e^{\beta_0+\beta_1 x_i}}{1+e^{\beta_0+\beta_1 x_i}}$$

(c) Using the modified function from part (b)iv, the likelihood function is the product of the probabilities of the observed independent outcomes:

$$L(\beta_0, \beta_1) = \prod_{i=1}^{n} p_i(x_i)^{y_i}(1-p_i(x_i))^{1-y_i} = \prod_{i=1}^{n} \left(\frac{e^{\beta_0+\beta_1 x_i}}{1+e^{\beta_0+\beta_1 x_i}}\right)^{y_i} \left(\frac{1}{1+e^{\beta_0+\beta_1 x_i}}\right)^{1-y_i}$$

(d) We can use Maximum Likelihood Estimation (MLE). By taking the natural logarithm of the likelihood function to obtain the log-likelihood $\log L(\beta_0, \beta_1)$, we can compute its partial derivatives with respect to β_0 and β_1. Setting these partial derivatives to 0 gives a system of equations. Solving this system (typically using numerical optimization algorithms like `optimise` in R) will yield the estimates $(\hat{\beta}_0, \hat{\beta}_1)$ that maximize the likelihood of observing our data.